

MIXED SIGNALS AND SENSORS

What is Amplifier in FMSS?

- **Amplifier** is a device or circuit that **takes a small electrical signal from a sensor or mixed-signal source and increases its strength** (voltage, current, or power) so it can be processed, transmitted, or recorded effectively.

IN SENSORS

- Many sensors (like temperature sensors, strain gauges, microphones, photodiodes) produce **very small signals** in the millivolt or microvolt range.
- An amplifier boosts these weak signals to a **usable level** for analog-to-digital converters (ADC) or controllers.

IN MIXED SIGNALS

- **Mixed signals** involve both **analog** (continuous) and **digital** signals in the same system.
- Amplifiers are used in the analog front-end to prepare sensor signals for digitization or further processing.

TYPES OF AMPLIFIERS

- **Voltage Amplifier** – Increases the voltage of a weak signal.
- **Current Amplifier** – Increases current for driving loads.
- **Operational Amplifier (Op-Amp)** – Common in signal conditioning for sensors.
- **Instrumentation Amplifier** – Specially designed for accurate, low-noise amplification of sensor outputs

INTRODUCTION TO OP-AMPS

Divisions of Electronics

Digital Electronics

All phases of electronics in which signals are represented in terms of finite number of digits. The distinguishing feature of digital electronics is the representation of varying physical quantities by means of discrete signals interpreted as numbers (usually in binary system).

Linear Electronics

All phases of electronics in which signals are represented by continuous or analog variables. Linear electronics circuit outputs varies directly with the changes in the input, which is why one of the most applications in the field of linear electronics is **AMPLIFICATION**.

A **SIGNAL** is a time-varying or spatial-varying quantity that carries information. It is a function of an independent variable such as time, distance, position, or temperature, etc.

A **SENSOR** is a device or instrument that detects and measures physical quantities, environmental conditions, or changes in its surroundings and converts this information into a signal suitable for processing.

FUNDAMENTALS OF MIXED SIGNALS AND SENSORS

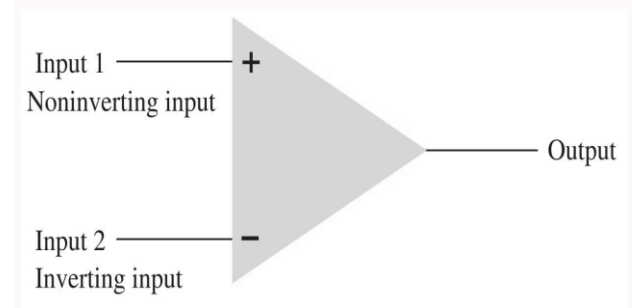
INPUT SIGNAL (STIMULI) → SENSOR (INDUCERS) → OUTPUT SIGNAL (AC OR DC SIGNAL) → AMPLIFIERS → PROCESSING CIRCUITS

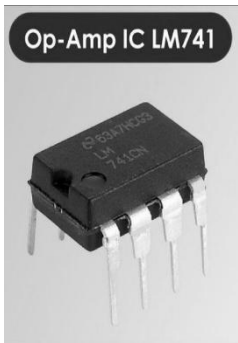
What is an Operational Amplifier?

Operational Amplifier

An operational amplifier, or op-amp, is a **very high gain** differential amplifier with **high input impedance** (typically few megaohms) and **low output impedance** (less than 100 ohms).

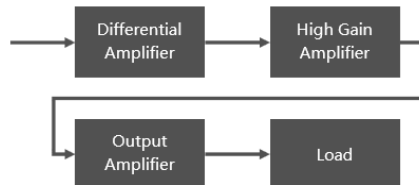
An op-amp contains a number of **differential amplifier stages** to achieve a very high voltage gain.





An **OPERATIONAL AMPLIFIER** (often abbreviated as "op-amp") is a high-gain, direct-current (DC) electronic amplifier with two inputs and a single output. It is a fundamental building block in analog electronic circuits and is widely used in a variety of applications due to its versatility and performance characteristics.

simplified block diagram



Of the many linear integrated circuits that are available, perhaps the **most widely used** (if not versatile) is the operational amplifier. It is capable of performing a **wide variety of linear and non-linear amplification and signal processing functions**.

Typical uses of the operational amplifier are to provide voltage amplitude changes (amplitude and polarity), oscillators, filter circuits, and many types of instrumentation circuits.

Typical Applications of Op-Amp

1. LINEAR AMPLIFIERS

- Non-inverting amplifier
- Inverting amplifier
- Voltage follower
- Summing amplifier
- Difference amplifier
- Instrumentation amplifier
- Differentiator
- Integrator

2. NON-LINEAR PROCESSING CIRCUITS

1. Precision rectifier
2. Peak detector
3. Comparator
4. Sample-and-Hold amplifier

3. WAVEFORM GENERATORS

- Phase-shift oscillators
- Twin-T oscillators

- Wien-bridge oscillators
- Square-wave generators

4. VOLTAGE AND CURRENT REGULATORS

- Series regulator
- Shunt regulator
- Current regulator

5. ACTIVE FILTERS

- Sallen-Key Network
- Multiple-Feedback Network
- State-Variable Filter
- Twin-T Filters

History of Operational Amplifiers



The term operational amplifier was originally applied to high-performance DC differential amplifiers that used *vacuum tubes*.

1948 - vacuum op-amp was capable of performing mathematical operations

But because of its size and power requirements, the vacuum tube soon gave way to the *transistor* ---> *integrated circuits* as improvements in technology were introduced.

1963 - The first commercially available op-amp

A702 were made by Fairchild

Semiconductor (designed by Bob Widlar).

1965 - A709 was also introduced by Fairchild

- National Semiconductor introduced **LM101**

1967 - Fairchild unveiled A741 which became the industry standard

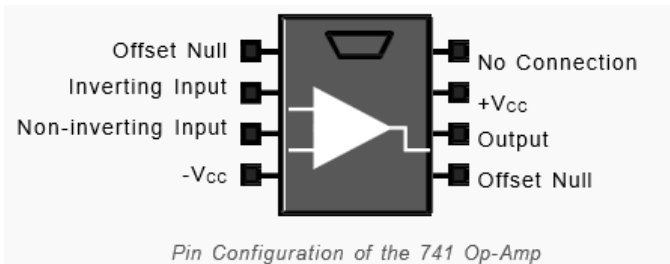
Other versions of the monolithic op-amps equivalent to A741 were created:





The 741 Op-Amp

- ▶ The 741 Op-Amp or operational amplifier is a DC-coupled high gain differential amplifier with external negative feedback.
- ▶ It consists of **8 pins** and It has only **one op-amp** inside.
- ▶ The main function of the 741 Op-Amp IC is to do the mathematical operations in various circuits. The LM741 Op-Amp IC can operate with a single or dual power supply voltage.



LESSON 2 CALCULATIONS NOT INCLUDED

Ideal Op-Amp Characteristics

The *ideal op-amp* would be expected to have the following ideal characteristics:

INFINITE OPEN-LOOP VOLTAGE GAIN

- This is the voltage gain of the op-amp without any external feedback and varies with frequency

INFINITE INPUT IMPEDANCE

- This is the internal resistance or impedance as seen at either the non-inverting input with respect to ground while the remaining input is grounded. This would show that the op-amp ideally does not require an input current.

ZERO OUTPUT IMPEDANCE

- This is the internal output resistance or impedance as seen from the output terminal with respect to ground. This would show that the op-amp ideally could efficiently transfer voltage signals from its output.

The *ideal op-amp* would be expected to have the following ideal characteristics:

INFINITE BANDWIDTH

- This is the frequency at which the open-loop voltage gain decreases to 1 or 0dB
- This is sometimes called the *unity gain frequency/bandwidth*. This shows that the op-amp ideally can amplify any frequency from DC to infinity.

INFINITE COMMON-MODE REJECTION RATIO

- This is the measure of the ability of the op-amp to reject common-mode signals that is simultaneously present at both inputs.

INFINITE SLEW RATE

- This is the time rate of change of the output voltage with the op-amp circuit having a closed-loop gain of 1. Ideally, the output voltage should exactly follow the input signal without distortion. For a rapidly changing input signal, such as square wave, the output should also resemble a square wave.

The *ideal op-amp* would be expected to have the following ideal characteristics:

ZERO INPUT OFFSET VOLTAGE

- This is the equivalent DC voltage that must be applied to one of the input terminals to produce a zero output voltage while the other input is grounded.

ZERO INPUT BIAS CURRENT

- This is the average of the two DC currents that flow into the inverting and non-inverting inputs and is the dc current required by the inputs of the amplifier to properly operate the first stage.

ZERO INPUT OFFSET CURRENT

- This is the difference of the two input bias currents whose polarity can either be positive or negative.

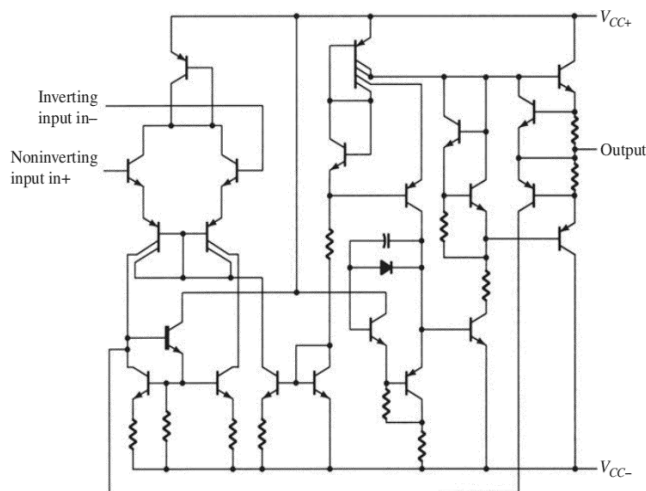
Parameters	Symbol	Ideal Value
Open-loop Voltage Gain	A_{OL}	Infinite
Input Impedance	R_{IN}	Infinite
Output Impedance	R_{OUT}	Zero
Bandwidth	f_{unity}	Infinite
Common-Mode Rejection Ratio	CMRR	Infinite
Slew Rate	slew rate	Infinite
Input Offset Voltage	$V_{IN(off)}$	Zero
Input Bias Current	$I_{IN(bias)}$	Zero
Input Offset Current	$I_{IN(off)}$	Zero

Parameters	Symbol	Ideal Value	LM741C
Open-loop Voltage Gain	A_{OL}	Infinite	200,000
Input Impedance	R_{IN}	Infinite	2 M Ω
Output Impedance	R_{OUT}	Zero	75 Ω
Bandwidth	f_{unity}	Infinite	100 MHz
Common-Mode Rejection Ratio	CMRR	Infinite	90 dB
Slew Rate	slew rate	Infinite	0.5 V/ μ s
Input Offset Voltage	$V_{IN(off)}$	Zero	2 mV
Input Bias Current	$I_{IN(bias)}$	Zero	80 nA
Input Offset Current	$I_{IN(off)}$	Zero	20 nA

The 741 Op-Amp

- The 741 Op-Amp or operational amplifier is a DC-coupled high gain differential amplifier with external negative feedback.
- It consists of **8 pins** and It has only **one op-amp** inside.
- The main function of the 741 Op-Amp IC is to do the mathematical operations in various circuits. The LM741 Op-Amp IC can operate with a single or dual power supply voltage.

Schematic Diagram of a Typical 741 Op-Amp



741 Op-Amp Specifications

Absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

	uA741	UNIT
Supply voltage V_{CC+}	22	V
Supply voltage V_{CC-}	-22	V
Differential input voltage	± 30	V
Input voltage any input	± 15	V
Voltage between either offset null terminal (N1/N2) and V_{CC-}	± 0.5	V
Duration of output short-circuit	unlimited	
Continuous total power dissipation at (or below) 25°C free-air temperature	500	mW
Operating free-air temperature range	-40 to 85	°C
Storage temperature range	-65 to 150	°C
Lead temperature 1.6 mm (1/16 in.) from case for 60 seconds	300	°C
Lead temperature 1.6 mm (1/16 in.) from case for 10 seconds	260	°C

The 741 Op-Amp Maximum Ratings

These are the maximum values that the op-amp can safely handle without the possibility of destruction. Under no circumstances should they be equal or exceeded.

SUPPLY VOLTAGE (+ V_{CC} , - V_{EE})

- This is the maximum positive and negative supply voltages that can be supplied to the op-amp.
- For the 741 op-amp, the max supply voltage is **± 22 V**.

DIFFERENTIAL INPUT VOLTAGE (V_D)

- This is the maximum voltage that can be applied between the two inputs.
- For the 741 op-amp, this is typically **± 30 V**.

COMMON-MODE VOLTAGE / INPUT VOLTAGE (V_{ICM})

- This is the maximum input voltage that can be simultaneously applied between both inputs and ground. In general, this is equivalent to the positive supply voltage.
- For the 741 op-amp, this is typically **± 17 V**.

These are the maximum values that the op-amp can safely handle without the possibility of destruction. Under no circumstances should they be equal or exceeded.

INTERNAL POWER DISSIPATION (P_D)

- This is the maximum power that the op-amp is capable of dissipating at a specified ambient temperature. For higher temperatures, this must be de-rated or reduced accordingly.
- For the 741 op-amp, this is typically **500 mW**.

OPERATING TEMPERATURE (T_A)

- This is the ambient temperature (25 °C) for the which the op-amp will operate within the

manufacturer's specifications. The military-grade version has a wider temperature range than the commercial version.

- For the 741 op-amp, this is typically **-40 °C to 85 °C**.

OUTPUT SHORT-CIRCUIT DURATION

- This is the time duration that the op-amp's output terminal can be short-circuited to ground or the supply voltage without damage.
- For the 741 op-amp, the output can be short-circuited **indefinitely**.

UNIT 2.1

SENSORS, TRANSDUCERS AND CONTROL DEVICES

Input Block

- It accepts the data signals from the external world into the system.
- The data signals may be directed to any number of destinations.

Sensor

- A sensor is a device that measures a physical quantity.

II. SENSORS

1. Light Sensitive Sensor

- An **electromagnetic radiation detector**.
- Uses the **ultraviolet to far infrared spectral range**.

2. Temperature Sensor

- Provides **temperature feedback** to a system controller.
- Allows the controller to **make decisions**.

3. Line-Tracking Sensor

- Can detect:
 - **White lines in black**
 - **Black lines in white**
- Uses the **TTL signal**.

4. Ultrasonic Sensor

- Uses **sonar** to determine the **distance from an object**.

5. Motion Sensor

- An electronic device used for **motion detection**.

6. Gas Sensors

- Widely used to detect **gas leakages**.
- Can detect:
 - LPG
 - Butane
 - Propane

- Methane
- Alcohol
- Hydrogen
- Smoke

- Used in **houses and factories**.

7. Hall Sensor

- An **omni-polar magnet sensor**.
- Used to detect **magnetic objects**.

8. Color Sensor

- A complete **color detector**.
- Can detect and measure a **nearly limitless range of visible colors**.

9. Digital Tilt Sensor

- A digital **tilt switch**.
- Has two modules:
 - **Mercury-based**
 - **Ball-based**

10. Triple Axis Accelerometer

- An **analog voltage output sensor**.
- Detects the **acceleration of a moving object**.

11. Analog Sound Sensor

- Used for detecting **loud sounds** in an ambient environment.

12. Voice Recognition

- A technology through which **sounds, words, or phrases spoken by humans** are converted into **electrical signals**.

13. Vibration Sensor

- A **digital plug** that plays sensor blocks.

14. Flame Sensor

- Can be used to detect **fire** or other wavelengths at **760–1100 nm light**.

15. Capacitive Touch Sensor

- When an object with **capacitive characteristics** comes close to a capacitive

touch sensor, it acts as **another capacitor** due to its **dielectric nature**.

16. Joystick Module

- Connects to **two analog outputs** based on commands:
 - **X control**
 - **Y control**

17. Biometric Sensors

Methods for biometric authentication include:

- **Iris scans**
- **Fingerprint sensors**
- **Face recognition**

18. Radio-Frequency Identification (RFID)

- Technology based on the **response of a tag to radio frequency signals**.

III. DATAPATH COMMUNICATION

Communication Media

- The physical entities that enable **communication**.
- The probability of communication errors can be computed using **communication theory techniques**.

Types of Communication Media:

1. **Wireless media**
2. **Optical media**
3. **Wires**

IV. MEMORY BLOCK

Types of Memory

ROM – Read-Only Memory

- Contains **embedded instructions**.
- Can be **flash-programmable**.

RAM – Random Access Memory

- **Volatile** memory.

Types/Examples under RAM:

- **Processor registers**
- **Working memory (Main Memory RAM)**
- **Buffer memory**
 - Caches
 - Buffers
 - Scratchpad memory

Sensor/Device	Main Function
Light Sensitive Sensor	Detects electromagnetic radiation
Temperature Sensor	Provides temperature feedback
Line-Tracking Sensor	Detects lines
Ultrasonic Sensor	Determines distance
Motion Sensor	Detects motion
Gas Sensor	Detects gas leakages
Hall Sensor	Detects magnetic objects
Color Sensor	Detects visible colors
Digital Tilt Sensor	Detects tilt
Triple Axis Accelerometer	Detects acceleration
Analog Sound Sensor	Detects loud sounds
Voice Recognition	Converts human speech into electrical signals
Vibration Sensor	Plays sensor blocks
Flame Sensor	Detects fire
Capacitive Touch Sensor	Detects capacitive objects/touch
Joystick Module	X and Y control
Biometric Sensor	Authentication
RFID	Responds to radio frequency signals

